Shantanu

ECE Undergraduate | Embedded Systems Engineer (ESP32, Sensors, IoT)
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Linkedin | **Portfolio** | **GitHub**

Education

- **Thapar Institute of Engineering and Technology** — Patiala, Punjab
  Bachelor of Technology - Electronics and Communication Engineering — **August 2025 - Present**
- **Thapar Institute of Engineering and Technology** — Patiala, Punjab
  Minor Degree - Computer Science and Engineering — **August 2026 - December 2027**

Skills Summary

- **Embedded Systems**: ESP32, Arduino, I2C, UART, SPI, MQTT, Real-time Processing
- **Programming**: C, C++ (Arduino), Python, JavaScript
- **Hardware / Electronics**: Sensors, Electromagnets, Power Electronics, GPS Modules, OLED Displays
- **Tools / Platforms**: Git, GitHub, KiCad, Arduino IDE, ESP-IDF, Vercel

Selected Projects & Open-Source Contribution

- **Adafruit CircuitPython - Wi-Fi API** — **C, CircuitPython**
  Open-source contribution in C: Implemented Raspberry Pi support for `wifi.radio.ap_info`; approach credited in the merged upstream PR. Initial PR #11625.
- **VIGIL-FUSE** — **Python, IoT Security**
  Built a Python CLI for embedded-firmware security analysis with 7 native rules and evidence-backed JSON/SARIF findings.
- **Zephyr Station** — **ESP32, C++**
  Integrated 7 sensor modules into an ESP32 monitor with OLED, SD logging, alerts, and a live dashboard.
- **Jolt Locator** — **ESP32, KiCad**
  Designed a custom PCB for an ESP32 GPS/compass navigator with OLED guidance and real-time heading/distance.
- **Energy Consumption Analysis** — **Python, PCA, K-Means**
  Clustered 200 synthetic household profiles over 365 days with PCA/K-Means; recovered 4 usage archetypes (ARI 0.813).

Experience

- **Ragastra MV Lev** — Patiala, Punjab
  Hardware & Controls Engineer — **September 2025 - Present**
  - **Electromagnetic System Design**: Designed and tuned custom electromagnets for magnetic levitation prototype with optimized coil parameters.
  - **Power Electronics**: Designed and debugged SMPS-based power circuitry for inductive electromagnetic loads.
  - **Motor Control Systems**: Implemented Arduino-based synchronized dual stepper motor control with real-time actuation.
- **Grosity** — Patiala, Punjab
  Early-Stage Founding Engineer — **October 2025 - February 2026**
  - **Web Development**: Developed and deployed company website using AI-assisted development workflows (Claude, GitHub Copilot, Kiro IDE).
  - **Autonomous Systems R&D**: Contributed to embedded electronics and control logic for autonomous payload-delivery drone systems.
  - **Product Strategy**: Aligned engineering decisions with B2B and D2C product models and business strategy.

Ongoing Research

- **GHFCVs - Reverse Engineering Study** — **December 2025 - Present**
  Literature review and comparative analysis of DC-DC converter topologies with hands-on teardown of commercial GHFCV systems to understand power electronics architecture and operational principles.

Certifications & Achievements

- **Internet of Things** - MIT Professional Education x Santander Open Academy
- **Google: Artificial Intelligence and Productivity** - Google x Santander Open Academy
- **National Finalist** - E Summit Ideastorm 2026, IIT Roorkee